Initial Project Proposal

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1. Project Title

Smart Visitor Pass Management System

2. Project Description

The Smart Visitor Pass Management System is a web-based application that digitizes the process of registering, tracking, and monitoring visitors at an organization's premises. Instead of writing visitor details in a paper register, the system allows security staff to register a visitor's details digitally, automatically generates a unique QR-coded pass for the visitor, and uses QR code scanning to record the visitor's entry and exit time. The purpose of the system is to make visitor management faster, more accurate, and easier to search, while giving administrators a live view of everyone currently on the premises.

3. Problem Statement

Most institutions still manage visitors using a manual, paper-based register at the security desk. This approach is slow, prone to human error, and makes it very difficult to search past visitor records. There is no real-time way to know how many visitors are currently inside the premises, and exit times are often not recorded at all, leaving visit records incomplete. There is also no unique verifiable pass for each visitor, making identity confirmation fully dependent on the security guard's manual judgement. This project aims to solve these problems by replacing the manual register with a digital, QR-code based visitor management system.

4. Project Objectives

- To develop a web-based system for registering visitor details digitally.
- To automatically generate a unique QR-coded pass for every visitor.
- To enable QR-code scanning at the gate for recording entry and exit times.
- To provide a live dashboard showing visitors currently inside the premises.
- To maintain a searchable history of all visitor records by name, host, or date.
- To reduce manual record-keeping errors and improve data accuracy.

5. Target Users

- Security Staff — register visitors and scan passes at the gate.
- Administrators — monitor the dashboard and review visitor history.
- Visitors — receive and use the QR-coded pass during their visit.

6. Main System Features

- Secure login for Admin and Security Staff (role-based access).
- Visitor registration (name, phone, email, purpose, host details).
- Automatic QR-coded visitor pass generation.
- QR code scanning for entry and exit tracking.
- Live dashboard of visitor statistics and current occupancy.
- Searchable visitor history with date-range filters.

7. Proposed Technology Stack

Frontend: HTML, CSS, JavaScript
Backend: PHP
Database: MySQL

8. Expected Outcome

By the end of the project, a fully functional Visitor Pass Management System is expected to be delivered, capable of registering visitors, generating QR-coded passes, tracking entry and exit through QR scanning, and displaying live visitor statistics along with a searchable history. This will eliminate the need for a manual paper register and give administrators accurate, real-time visibility into visitor activity on the premises.